

Xiwei Wang

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Education

University of Waterloo

09/2025 - 12/2026 (Expected)

Master of Engineering in Electrical and Computer Engineering (Software Engineering)

Yanshan University

09/2021 - 06/2025

Bachelor of Electronic Science and Technology, GPA: 86/100

TECHNICAL SKILLS

- **Languages:** Java, TypeScript, JavaScript, Python, C/C++, SQL, HTML/CSS
- **Frontend & Mobile:** React, React Native, Flutter
- **Backend & Cloud:** Spring Boot, Node.js/Express, Django, REST APIs, Azure, Google Cloud, Docker, CI/CD
- **Data & AI:** PostgreSQL, MySQL, MongoDB, Redis, Qwen/Gemini, Stable Diffusion

PROFESSIONAL EXPERIENCE

IPMD, Inc.

05/2026 - 08/2026

Software Engineering Intern

California, United States - Remote

- Built and deployed an end-to-end AI image-generation platform on Microsoft Azure, integrating React/TypeScript, Node.js/Express, Qwen/Gemini, and an internal Stable Diffusion 1.5 inference pipeline to support multimodal artwork generation.
- Containerized the React frontend, Node.js backend, Qwen service, and Stable Diffusion inference service with Docker, enabling reproducible deployment across multiple AI and application services.
- Streamlined CI/CD workflows and coordinated six cross-functional workstreams across engineering, design, and product to align implementation priorities and deliver the MVP.

FoloBotics

01/2026 - 04/2026

Full-Stack Developer

Ontario, Canada

- Developed scalable Java/Spring Boot backend services and REST APIs for user, task, recommendation, and warehouse workflows consumed by Flutter clients.
- Refactored Java/Spring Boot API and data-access layers with a multi-threaded architecture, mitigating race conditions and improving thread-safe operations across PostgreSQL and Redis, reducing average response time by 25%.

SELEST Security

09/2025 - 12/2025

Full-Stack Developer

Ontario, Canada

- Built and shipped a multi-role React Native application using TypeScript and JavaScript, Node.js/Express, MongoDB, and Google Cloud, supporting production releases to the App Store and Google Play.
- Implemented 5+ booking and shift-management workflows through REST API integration, ensuring reliable state synchronization and data consistency across user roles.

Harbin Institute of Technology Robot Technology Research Institute

05/2025 - 08/2025

Industrial Vision Platform Development Intern

China

- Built a real-time C++/Qt computer-vision application on Linux for teacup defect detection that processed up to 600 images per second from live camera feeds, optimizing memory management and integrating application modules.
- Developed a Python/Django REST backend for detection logging, parameter synchronization, and offline frame analysis, connecting vision outputs to structured application services.

Sinopec Group

06/2024 - 12/2024

Application Intern at the Information Center

China

- Developed a real-time 2D warehouse visualization component using React and Canvas/SVG, rendering 5,000+ storage locations with live status highlighting and heatmap analytics while maintaining smooth UI performance.
- Built real-time data synchronization using WebSocket (Socket.io), streaming PDA scan events and AGV position updates to the React dashboard with under 80 ms latency.

ADDITIONAL INFORMATION

Languages: English (fluent), Chinese (native), French (basic)